



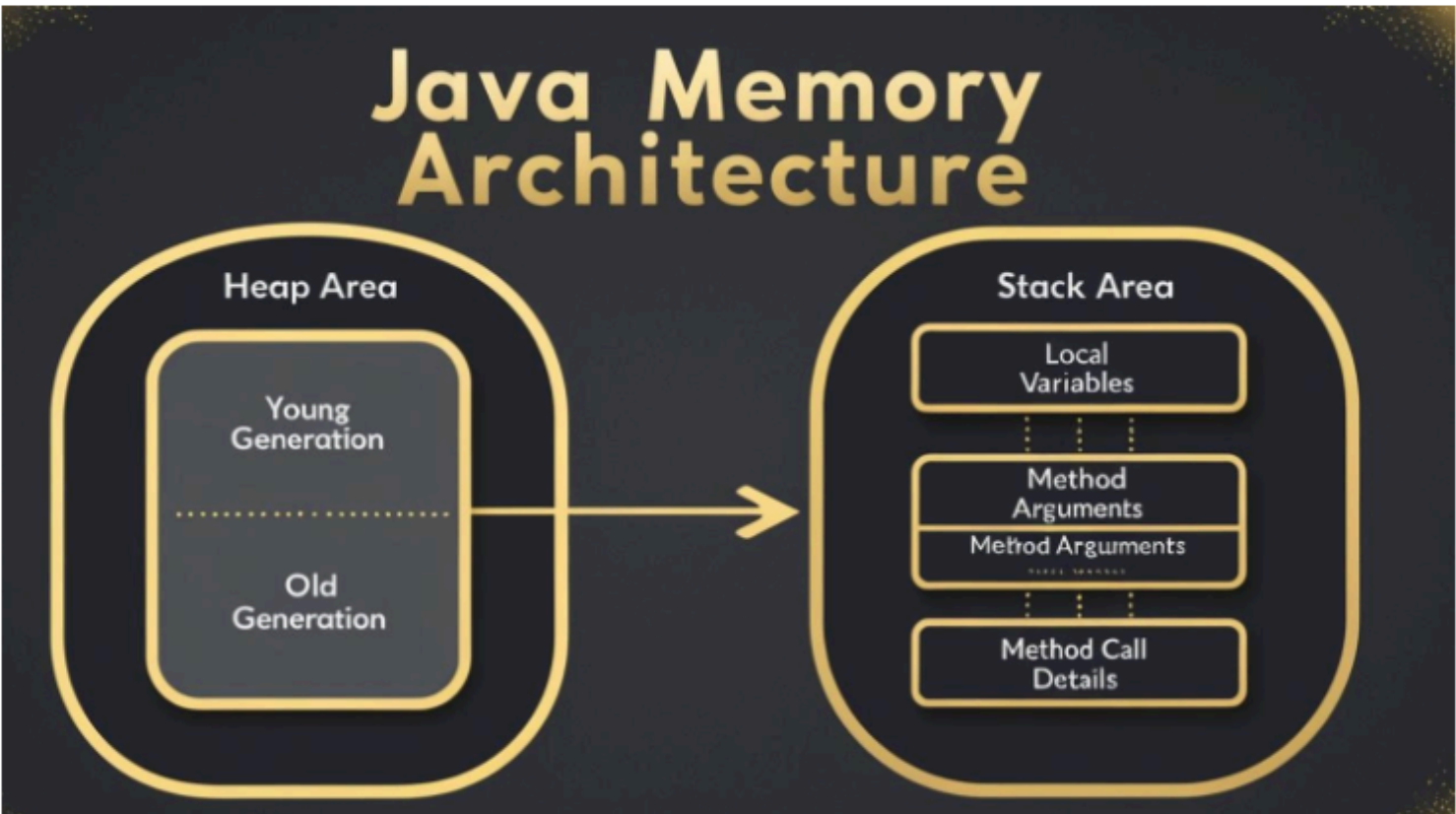
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JAVA MEMORY ARCHITECTURE

In Java, memory is divided into two main areas:

- **Heap Area**
- **Stack Area**



Heap Area

- Managed by the garbage collector.
- Used to store objects.
- Here memory allocation is dynamic.
- Memory grows and shrinks automatically.
- Objects in the heap are allocated dynamically.

Internally, the heap memory is further divided into two parts:

Young Generation	Old Generation
▪ New objects are allocated here. ▪ Short-lived objects are stored here.	Long-lived objects are stored here.

Why are there two areas inside the heap:

- It is a memory management pattern followed by the JVM to manage memory efficiently.
- In the **Young Generation**, the JVM frequently cleans up short-lived objects to free up memory space.

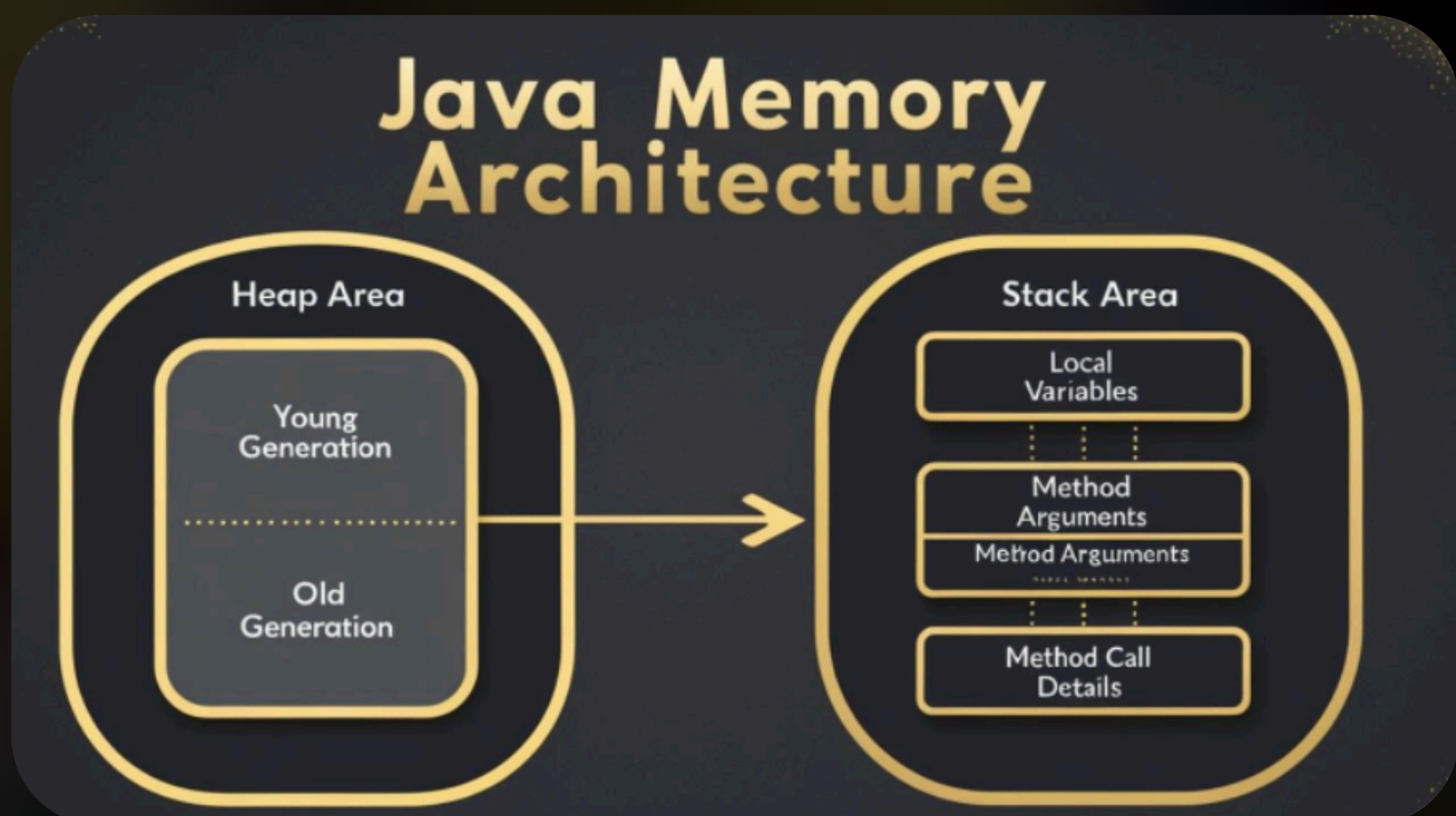
Note: Objects are stored in the **Heap Area**, while references are stored in the **Stack Area**.



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JAVA MEMORY ARCHITECTURE



Stack Area

- Memory allocation is not dynamic (it has a fixed size).
- Holds information related to:
 - Local variables
 - Method arguments
 - Method call details

Other Memory Areas

Method Area (PermGen) – Before Java 8

- Used to store class-level data.
- Stores:
 - Class definitions
 - Metadata about classes (e.g., class file information)
 - Data about methods

After Java 8: Metaspace Area

- Uses memory from the heap area.
- Memory allocation is now dynamic (previously static).

Native Method Stack

- Stores information about **native methods** used in the Java application.

PC Register

- Stores information about the **current execution address**.



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PURPOSE OF THIS DOCUMENT

- **Use this PDF as a quick last-minute revision guide only.**
- **For more details, please perform R&D and gain a deeper understanding of how memory architecture works.**
- **Once the interview series is complete, we will cover this topic in detail.**



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